

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Basic Concepts on Software Engineering



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.

Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Can Do Statements:

- I can give brief information about what software engineering is.
- I can identify which are the main concepts in software engineering.
- I can explain briefly the importance of software engineering.

Ask your partner:

- What is computer software?
- Do you think there is a difference between computer software and computer program?





Let's Explore!

Welcome to Introduction to Software* Engineering

*"Software" refers to the group of programs used to operate computers.

```
Exec {
  path => ["/bin", "/sbin", "/usr/bin", "/usr/sbin"],
}

include apache

a2mod { "Enable proxy mod":
  name => "proxy",
  ensure => "present"
}

a2mod { "Enable proxy_http mod":
  name => "proxy_http",
  ensure => "present"
}

apache::vhost::proxy { "my-site":
  servername => "my-site.com",
  port => 80,
  dest => "http://localhost:9015",
  vhost_name => "my-site"
}
```

Instructions: Read the text below.

Good morning, class. Welcome to Introduction to Software Engineering. I'm Mr. Hardiman.

So, What is software engineering? I think you already know, but we need to review some essential aspects that define software engineering. Future programmers need to know that software engineering is the implementation of engineering principles for software development. One of the main objectives of software engineering is to use skills, techniques and engineering tools for software production. Another important objective is to design methods and procedures that engineers use to produce high-quality and easy-to-use software.

There are some important concepts that software engineers need to take into account, but they are not difficult to remember.

The first concept is the interface. It is the most important concept in software engineering because a good piece of software is the model of a real or imaginary system. We also have conventions and templates that enable automation in software design. Additionally, layering is the programming of different separate components that interact in a hierarchical or sequential way. Finally, when coding, engineers have to take into account the algorithmic complexity; the code that they write needs to be clean, compact, and readable.

As you all can see, software engineering is an important field of engineering in the modern world. It helps companies minimize software cost and handle big projects. It also helps them increase productivity, guarantee the safety of sensitive data, and reduce the complexity or difficulty of tasks.

**Let's Practice!**

We are in the same engineering class.

Instructions: Read the statement. Select all the correct answers.

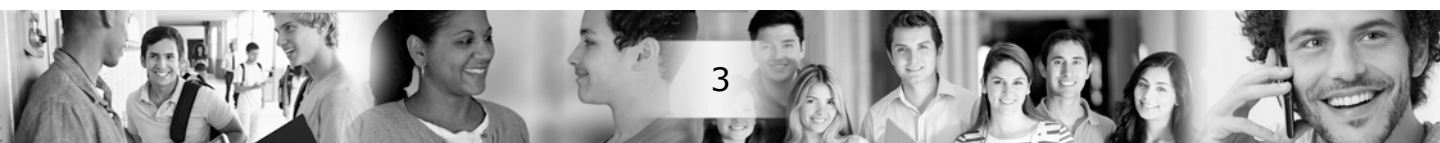
1. Software engineering is
 - a. the implementation of engineering principles for software production.
 - b. a new science for software distribution.
 - c. the design of methods to produce software.
 - d. a class only for software designers.
 - e. a field of engineering that minimizes costs for customers.

**Let's Practice!**

Learning What the Main Concepts of Software Engineering Are

Instructions: Put the words from the box into the gaps. Practice saying the sentences aloud. Check your pronunciation with your instructor. Use: and, but.

1. Interfaces _____ layering are the most important concepts in software engineering.
2. Algorithms are difficult _____ let's see more information about it.
3. Convention _____ templates are important patterns to generate codes.





Extension Activity

Make a concept map about software engineering.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about what software engineering is.
- Describe what software engineering is.
- Draw a concept map describing what the main concepts of software engineering are.
- Share your concept map with the class and write more concepts if it is necessary.

Self-Assessment

Yes

No

- I find information about what software engineering is. ☐ Yes ☐ No
- I understand what software engineering is. ☐ Yes ☐ No
- I draw a concept map describing what the main concepts of software engineering are. ☐ Yes ☐ No
- I share my concept map with the class. ☐ Yes ☐ No



Target Words

- engineering** (noun): the use of scientific and mathematical values to any process for the plan, production and task of an expected result
e.g. We need to review some essential aspects that define software engineering.
- layering** (noun): programming into segments or parts of a bigger component
e.g. Layering is the programming of different separate components that interact in a hierarchical or sequential way.
- handle** (verb): to manage; to be able to deal with something
e.g. It helps companies minimize software cost and handle big projects.
- enable** (verb): to make something possible
e.g. We also have conventions and templates that enable automation in software design.
- readable** (adjective): easy to read and understand
e.g. The code that they write needs to be clean, compact, and readable.



Learning Points

Simple Present Tense - Verb to Be in Affirmative Form

You use the **simple present** tense to talk about general actions, routines, and the general state or habits of a subject. The **verb to be** is one of the most important verbs in English because it has different functions.

Use the **simple present** tense of the **verb to be** using the following conjugation (**am, is, are**):

I am a software engineer.
You are an intelligent designer.
He is an expert in software application.
She is a good organizer of layering.
It is a good way to organize virtual documents.
We are designers of software.
They are famous for creating a financial software.

You can also use contractions to abbreviate the subject and the verb as follows:

I am	I'm
You are	You're
He is	He's
She is	She's
It is	It's
They are	They're
We are	We're

Remember, when you refer to a *third person*, you can use the corresponding personal pronoun or you can mention the specific subject. No matter which one you use, the conjugation is the same:

They are software engineers.
Mary and Adam are software engineers.

It is an application to design.
Software engineering is an application to design

Examples

- *There are some important concepts that software engineers need to take into account, but they are not difficult to remember.*
- *Future programmers need to know that software engineering is the implementation of engineering principles for software development.*
- *Welcome to Introduction to Software Engineering. I'm Mr. Hardiman.*





Learning Points

What Is Software Engineering?

Software engineering integrates different methods to design create, organize and develop computer programs, applications, and data, among others. It also helps to organize all kind of information and documents according to the purpose of the institution or subject who is going to use it.

Software engineering is 66 years old and it was essentially created to apply engineering principles to software production. Software is the variable and intangible part of a computer.

So when you talk about software you can refer to programs of a particular interest or operating systems; some examples of programs of particular interest are: Paint and the image editing programs; you can also find Linux or Windows as examples of operating systems.

Software engineering implies different actions to successfully develop any kind of software:

- Research
- Design
- Develop
- Implement
- Test
- Apply
- Documentation
- Organization
- Development
- Operation
- Maintenance

This process helps to produce software for multiple purposes, such as business, industry, financial services, scientific data and some others. All these steps in software engineering need to be consistent and efficient to integrate a functional software.